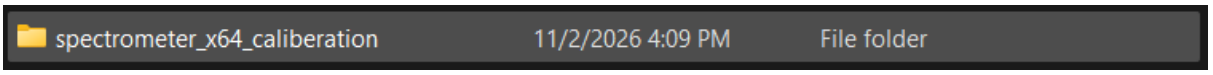
The thing you will need will most likely be in this folder:



To begin the calibration process, you have to first have the optic paths set up. Make sure that the light can correctly go into the monochromator and out into the fiber directed into the USB-spectrometer.

To do that you need to make both entrance and exit open to 500 micrometers and check if the spectrometer is getting any signal and the light source is on.

If all is working properly, you will need the maximum and minimum of turret steps range. For example : -250000 ~ 750000 for TRIAX 550
-0 ~ 32000 for TRIAX 320

it should be changed here:

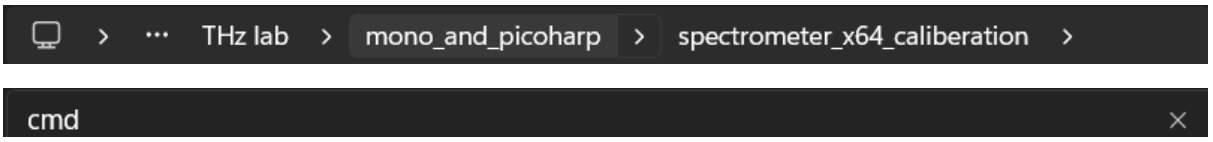
```
135         # Define scan sequence (e.g., 0 to 550000 in steps of 5000)
136         # Make sure to change this according to monochromater
137         scan_steps = list(range(0, 25000, 100))
```

and here for wanted range:

```
204             if 11000 >= s >= 5500:  # Primary diffraction order filter
```

The whole process requires more than one run of the code, first for determining the range for 1st order of wavelengths, other for optimization.

To run the code, simply type cmd like so.



you should then open up a terminal that had dir to the current folder.

Then you have to check the python version, any versions from 3.9 - 3.12 should work properly.
To check, just type

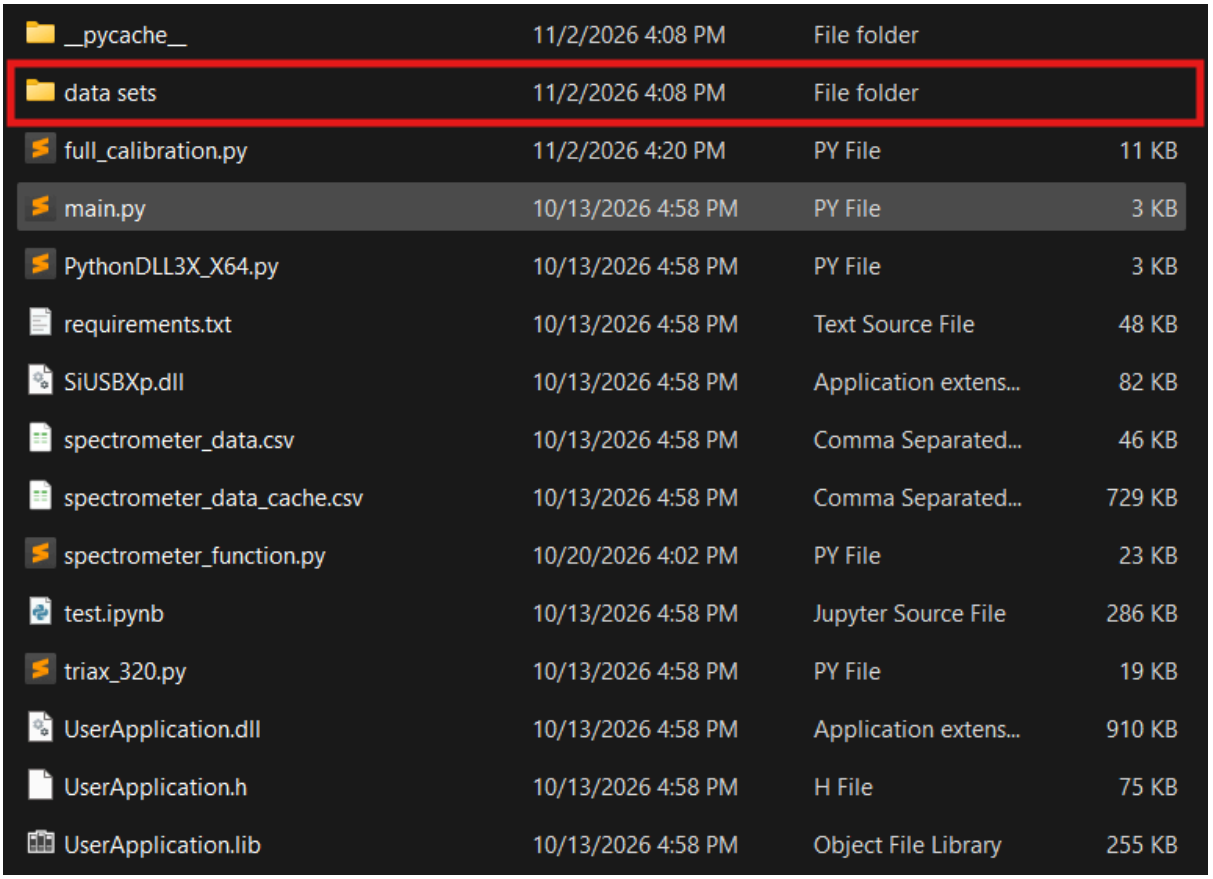
```
py --version
```

then type

```
py calibration.py
```

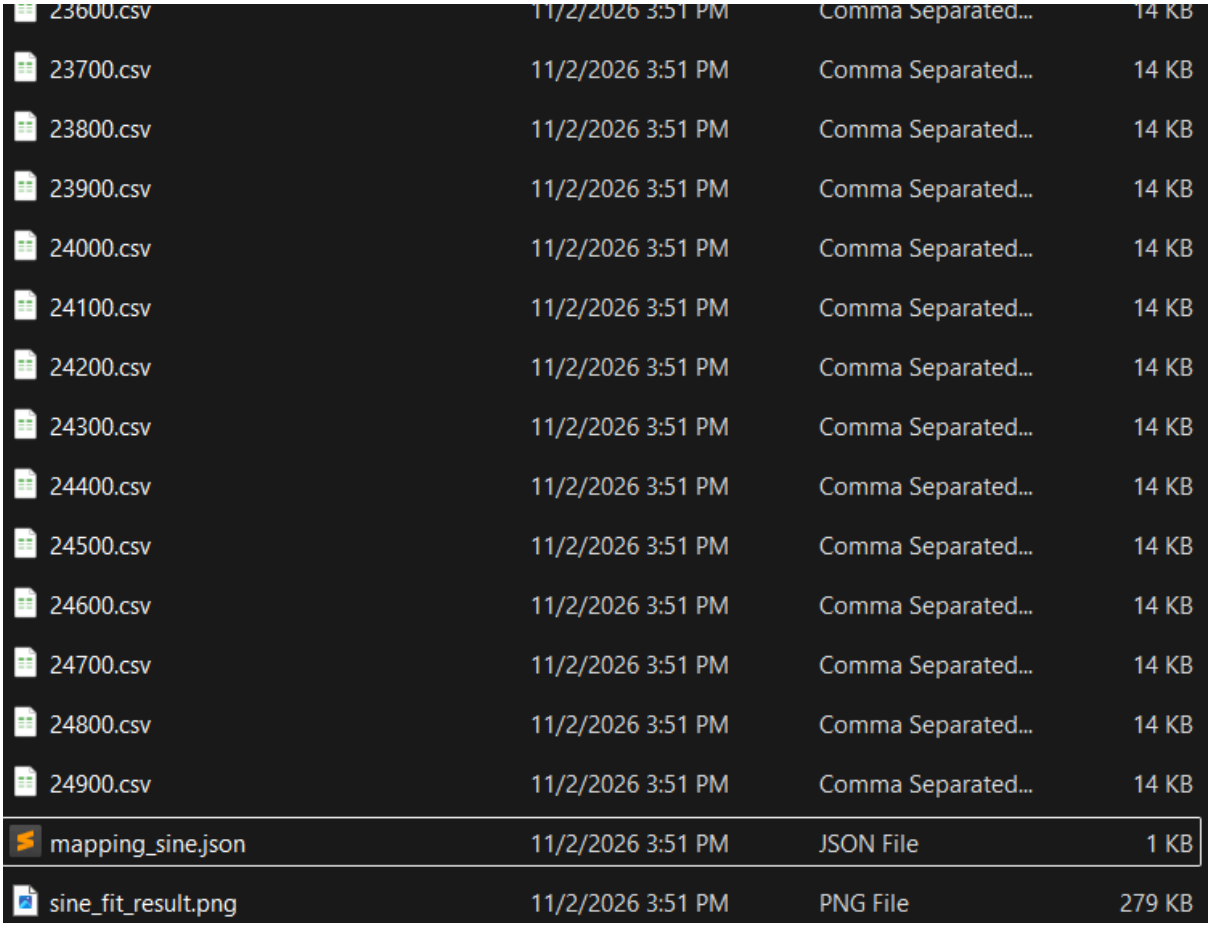
Upon finishing, a graph should show up and a list of data should appear at the bottom of the terminal, showing the result of fitting. Make sure if the R-squared value is acceptable.

All the recorded data should be in data sets folder



__pycache__	11/2/2026 4:08 PM	File folder	
data sets	11/2/2026 4:08 PM	File folder	
full_calibration.py	11/2/2026 4:20 PM	PY File	11 KB
main.py	10/13/2026 4:58 PM	PY File	3 KB
PythonDLL3X_X64.py	10/13/2026 4:58 PM	PY File	3 KB
requirements.txt	10/13/2026 4:58 PM	Text Source File	48 KB
SiUSBXp.dll	10/13/2026 4:58 PM	Application extens...	82 KB
spectrometer_data.csv	10/13/2026 4:58 PM	Comma Separated...	46 KB
spectrometer_data_cache.csv	10/13/2026 4:58 PM	Comma Separated...	729 KB
spectrometer_function.py	10/20/2026 4:02 PM	PY File	23 KB
test.ipynb	10/13/2026 4:58 PM	Jupyter Source File	286 KB
triax_320.py	10/13/2026 4:58 PM	PY File	19 KB
UserApplication.dll	10/13/2026 4:58 PM	Application extens...	910 KB
UserApplication.h	10/13/2026 4:58 PM	H File	75 KB
UserApplication.lib	10/13/2026 4:58 PM	Object File Library	255 KB

inside the folder should be a file called “mapping_sine.json”



use this to replace the file in mapping.json in the mono_ and_picoharp will conclude the calibration process.

Don’t forget to double check if the calibration actually worked by checking with the main mono_and_picoharp application to see if the wavelength was aligned or not.

Make sure to also change the range in main programme as well.

like here:

```
1223            self.ui.start_mono_pos_label.setText(">=-250000")
1224            self.ui.end_mono_pos_label.setText("<=750000")
1225            #For triax 320
1226            #self.ui.start_mono_pos_label.setText(">=0")
1227            #self.ui.end_mono_pos_label.setText("<=32000")
```

```
2844    invalid_steps = [s for s in self.x_values if s < -250000 or s > 750000]
2845    #For triax 320
2846    #invalid_steps = [s for s in self.x_values if s < -0 or s > 32000]
```

and so on